

Task 1

a) We know :

$$p_1(t+1) = (1 - \varepsilon) p_1(t) \quad - (1)$$

$$p_2(t+1) = \varepsilon p_1(t) + p_2(t) \quad - (2)$$

$$(1) + (2)$$

$$p_1(t+1) + p_2(t+1) = p_1(t) + \varepsilon \overbrace{p_1(t) - \varepsilon p_1(t)} + p_2(t)$$

by deduction

$$p_1(t) + p_2(t) = p_1(0) + p_2(0) \quad (\text{const})$$

b) $\lambda_1 = 1 \quad , \quad \lambda_2 = 1 - \varepsilon$

$$M = \begin{pmatrix} 1 - \varepsilon & 0 \\ \varepsilon & 1 \end{pmatrix}$$

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$1 - \varepsilon \quad 0$$

$$0 \quad 1 - \varepsilon$$

$$\begin{bmatrix} 0 & 0 \\ \varepsilon & \varepsilon \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$0x + 0y = 0$$

$$\xi x + \xi y = 0$$

$$\xi(x+y) = 0$$

$$x+y = 0$$

$$1 - 1$$

$$(a, -a)$$

$$\lambda_1 = 1 \quad \lambda_2 = 1 - e$$

$$1b) \quad \underset{v_1}{(0, 1)} \quad \underset{v_2}{(1, -1)}$$

1 c)

C_1, V_1

$$P(t) = \begin{pmatrix} p_1(t) \\ p_2(t) \end{pmatrix}$$

$$p_1(t) + p_2(t) = \frac{1}{2} + p_2(0)$$

$$p_2(0) + \frac{1}{2} = P(0) \begin{pmatrix} 1/2 \\ p_2(0) \end{pmatrix} = C_1 \vec{V}_1 + C_2 \vec{V}_2$$

$$\begin{pmatrix} 1/2 \\ p_2(0) \end{pmatrix} = \begin{bmatrix} 0 \\ C_1 \end{bmatrix} + \begin{bmatrix} C_2 \\ -C_2 \end{bmatrix}$$

$$C_2 = 1/2$$

$$C_1 - C_2 = p_2(0)$$

$$C_1 = p_2(0) + 1/2$$

$$p_2 = (1 - p_1)$$

$$c_2 = 1/2$$

$$c_1 = \frac{1}{2} + \frac{1}{2} = 1$$

$$\vec{p}(t=0) = \vec{v}_1 + \frac{\vec{v}_2}{2}$$

$$1d) \vec{p}(t) = c_1 \vec{v}_1 + c_2 \vec{v}_2$$

$$\vec{v}_1 + \frac{\vec{v}_2}{2}$$

$$\vec{p}(0) = c_1 \vec{v}_1 + c_2 \vec{v}_2$$

$$\vec{p}(t) = \begin{bmatrix} p_1 \\ p_2 \end{bmatrix}$$

$$= \begin{bmatrix} p_1(t) \end{bmatrix}$$

$$\begin{aligned}
 & \begin{bmatrix} p_1(t) \\ p_2(t) \end{bmatrix} \\
 &= \begin{bmatrix} (1-\varepsilon) p_1(t-1) \\ \varepsilon p_1(t-1) + p_2(t-1) \end{bmatrix} \\
 & \left[\begin{aligned} & \frac{(1-\varepsilon)^t}{2} \\ & \frac{t(1-\varepsilon)^t}{2} + \frac{t}{2} \end{aligned} \right]
 \end{aligned}$$

$$\begin{aligned}
 P(1) &= M P(0) = M v_1 + \frac{M v_2}{2} \\
 &= \lambda_1 v_1 + \frac{\lambda v_2}{2}
 \end{aligned}$$

$$\begin{aligned}
 P(2) &= M P(1) \\
 &= M \lambda_1 v_1 + \frac{M \lambda v_2}{2}
 \end{aligned}$$

$$P_0 = c_1 v_1 + c_2 v_2$$

$$P(1) = M v_1 c_1 + M v_2 c_2$$

$$\begin{aligned} P(t) &= (\lambda_1)^t c_1 + (\lambda_2)^t c_2 \\ &= (1)^t c_1 + (1-\varepsilon)^t c_2 \end{aligned}$$

$$1e) \quad P(t) = (\lambda_1)^t c_1 + (\lambda_2)^t c_2$$

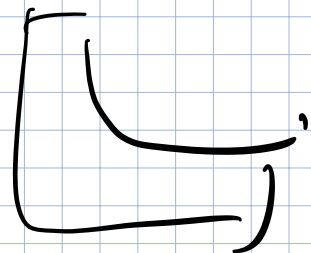
$$\begin{aligned} P(t) &= (1)^t c_1 + (1-\varepsilon)^t c_2 \\ &= \quad + \end{aligned}$$

$$\begin{aligned} p_1(t) &= (1-\varepsilon) p_1(t-1) \\ &= (1-\varepsilon)^t p_1(0) \end{aligned}$$

$$p_2(t) = \varepsilon p_1(t) + p_2(t)$$

Task 1(g)

① apply $\varepsilon = e^{(-t/t_0)}$



exponential
decay

$$P(t) = (1)^t C_1 + (1 - \varepsilon(t))^t C_2$$